Charis Stamouli

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RESEARCH INTERESTS

Control systems, learning for safe cyber-physical systems, statistical learning, layered control, model predictive control

EDUCATION

University of Pennsylvania (UPenn) 09/2020–present
Ph.D. in Electrical and Systems Engineering

- Advisor: George J. Pappas

National Technical University of Athens (NTUA) 09/2014–11/2019
Diploma in Electrical and Computer Engineering (MEng, five-year degree)

- Thesis: Multi-agent Formation Control based on Distributed Estimation with Prescribed Performance
- Advisor: Kostas J. Kyriakopoulos
- GPA: 9.50/10, rank in top 3% of class

PROFESSIONAL EXPERIENCE

Teaching Assistant at UPenn
CIS520: Machine Learning 01/2022–04/2022
ESE500: Linear Systems Theory 08/2021–12/2021

- Designed homework and exam sets, held office hours, and taught classes

Research Assistant at NTUA
Department of Mechanical Engineering, Control Systems Lab 11/2018–05/2020

- Conducted research in formation control of multi-agent systems based on distributed estimation

Robotics Intern at National Centre for Scientific Research "Demokritos"
Institute of Informatics and Telecommunications, Roboskel Lab 07/2018–08/2018

- Collected images of obstacles using a camera placed on a mobile robot
- Extended and implemented an algorithm for obstacle recognition during the autonomous motion of a vision-enabled mobile robot

Teaching Assistant at NTUA
Programming Techniques 02/2017–06/2017

- Assisted students in the lab sessions

HONORS AND AWARDS

American Control Conference Travel Grant 2026
for being the first author of "Policy Gradient Bounds in Multitask LQR"

Northeast Systems and Control Workshop Travel Grant 2025
for being selected to give an oral presentation

American Control Conference Best Student Paper Award 2024
for being the first author of "Structural Risk Minimization for Learning Nonlinear Dynamics"

American Control Conference Travel Grant 2024
for being a Best Student Paper Award finalist

Ganster Engineering Fellowship 2020
Department of Electrical and Systems Engineering, UPenn

Dean's Fellowship 2020
Department of Electrical and Systems Engineering, UPenn

NTUA Thomaidion Award 2019
for the publication "Robust Dynamic Average Consensus with Prescribed Performance"

Eurobank EFG Award 2014
for ranking first in my high school on the Nationwide University Entrance Examination

PUBLICATIONS

Conference papers

[1] L.F. Toso*, K. Fallah*, **C. Stamouli***, G. J. Pappas, and J. Anderson, "Multitask LQG Control: Performance and Generalization Bounds," IEEE Conference on Decision and Control (CDC), 2026. To appear.

[2] **C. Stamouli**, A. Tsiamis, and G. J. Pappas, "Layered Control of Partially Observed Stochastic Systems," IEEE Conference on Decision and Control (CDC), 2026. To appear.

[3] V. Tassopoulou*, **C. Stamouli***, H. Shou, G. J. Pappas, and C. Davatzikos, "Uncertainty-Calibrated Prediction of Randomly-Timed Biomarker Trajectories with Conformal Bands," Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.

[4] **C. Stamouli**, A. Tsiamis, M. Morari, and G. J. Pappas, "Layered Multirate Control of Constrained Linear Systems," IEEE Conference on Decision and Control (CDC), 2025.

[5] **C. Stamouli**, I. Ziemann, and G. J. Pappas, "Rate-Optimal Non-Asymptotics for the Quadratic Prediction Error Method," IEEE Conference on Decision and Control (CDC), 2024.

[6] **C. Stamouli**, L. Lindemann, and G. J. Pappas, "Recursively Feasible Shrinking-Horizon MPC in Dynamic Environments with Conformal Prediction Guarantees," Conference on Learning for Dynamics and Control (L4DC), PMLR, 2024.

[7] **C. Stamouli**, E. Chatzipantazis, and G. J. Pappas, "Structural Risk Minimization for Learning Nonlinear Dynamics," IEEE American Control Conference (ACC), 2024. **Best Student Paper Award.**

[8] **C. Stamouli**, A. Tsiamis, M. Morari, and G. J. Pappas, "Adaptive Stochastic MPC under Unknown Noise Distribution," Conference on Learning for Dynamics and Control (L4DC), PMLR, 2022.

[9] **C. J. Stamouli**, C. P. Bechlioulis, and K. J. Kyriakopoulos, "Robust Dynamic Average Consensus with Prescribed Performance," IEEE Conference on Decision and Control (CDC), 2019.

Journal papers

[1] **C. Stamouli***, L. F. Toso*, A. Tsiamis, G. J. Pappas, and J. Anderson, "Policy Gradient Bounds in Multitask LQR," IEEE Control Systems Letters, 2025. To be presented at ACC 2026.

[2] **C. J. Stamouli**, C. P. Bechlioulis, and K. J. Kyriakopoulos, "Robust Dynamic Average Consensus with Prescribed Transient and Steady State Performance," Automatica, 2022.

[3] **C. J. Stamouli**, C. P. Bechlioulis, and K. J. Kyriakopoulos, "Multi-Agent Formation Control Based on Distributed Estimation with Prescribed Performance," in IEEE Robotics and Automation Letters, 2020. Presented at ICRA 2020.

INVITED TALKS AND POSTERS (EXCLUDING CONFERENCE PRESENTATIONS)

Learning for Dynamics and Control 2026
Women in Control Luncheon, American Control Conference, New Orleans, LA

Uncertainty-Calibrated Prediction of Biomarker Trajectories (poster) 2026
AI in Medicine Symposium, University of Pennsylvania, Philadelphia, PA

Layering and Task Generalization in Control Architectures 2025–2026
Invited group seminars at University of Michigan (N. Ozay), MIT (G. Zardini), and Columbia University (J. Anderson)

Layered Multirate Control of Constrained Linear Systems 2025
Workshop on Control Architecture Theory, CDC, Rio de Janeiro, Brazil
Northeast Systems and Control Workshop, New York City, NY

Structural Risk Minimization for Learning Nonlinear Dynamics (poster), Emerging CORE Methods in Data Science Retreat (NSF HDR), University of California, San Diego, CA 2025

SKILLS

Programming	Python, C/C++, MATLAB (proficient), Java, Prolog, ML (past experience)
Languages	English (fluent), Spanish (advanced), Greek (native)

PROFESSIONAL SERVICE

Conference reviewer	CDC, L4DC, ACC, ECC, ICRA
Journal reviewer	Automatica, IEEE TAC, IEEE L-CSS
Invited session co-organizer	Control Architecture Theory, Conference on Decision and Control, 2026